COVID-19 Early Pandemic Data Analysis

Prepared by: Vishal Kumbhakar

Location: Jamhsedpur, Jharkhand India

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1. Project Overview & Objectives

This project involves an end-to-end exploratory data analysis on early-wave COVID-19 datasets (Jan-Aug 2020). The goal is to provide a clear understanding of the geographic distribution, temporal evolution, and demographic vulnerability patterns using Python for data engineering and Power BI for interactive visualization.

2. Datasets Description

- **covid.csv** Country-level snapshot (Aug 2020). Cases, Deaths, ISO codes.
- **covid_grouped.csv** Daily time-series (Jan-Jul 2020). New cases and recovery trends.
- **coviddeath.csv** US mortality data (Feb-Aug 2020) categorized by age and comorbidity.

3. Data Preparation & Cleaning

Using Python (Pandas), we handled missing values in 'NewCases' and 'NewDeaths' by imputing zeros. Population outliers (cruise ships) were handled separately. Date columns were standardized to ISO format to ensure seamless integration with the Power BI Time Intelligence functions.

4. Exploratory Data Analysis - Key Findings

* Global: Top cases were led by USA, Brazil, and India by August 2020.
* Temporal: Exponential growth was most pronounced in March-April 2020.
* US Mortality: Over 80% of deaths occurred in the 65+ age demographic.
* Comorbidity: Respiratory and Circulatory diseases were the primary contributing conditions.

5. Power BI Dashboard Architecture

The dashboard uses a star schema where the grouped time-series acts as the fact table. Key DAX measures include 'Global Fatality Rate (%)' and 'Positivity Rate'. The layout follows a Z-pattern, prioritizing high-level KPIs at the top followed by geospatial and demographic drill-downs.

Conclusion

The analysis confirms that early pandemic risk was heavily stratified by age and pre-existing health conditions. This documentation serves as a blueprint for the final Power BI interactive report.